Summary in Graph

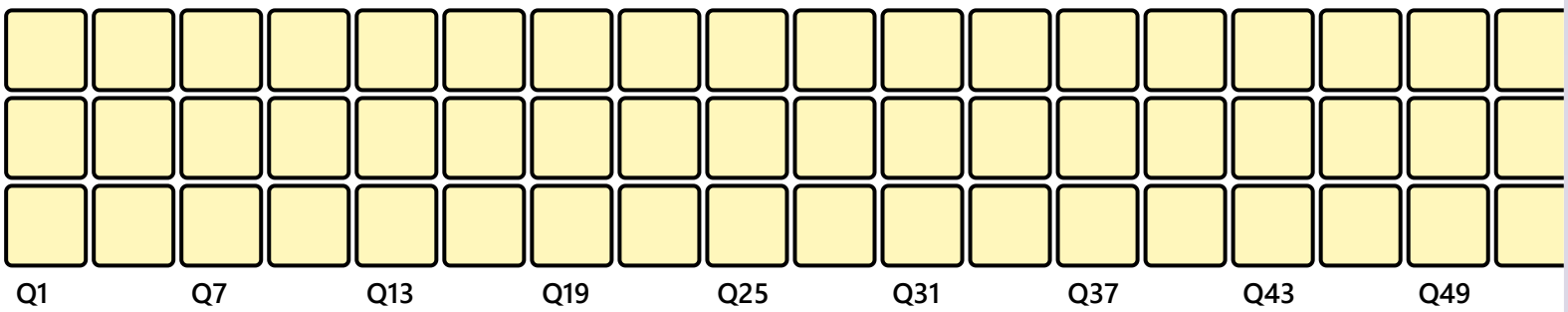
Exam Summary (GATE Overflow CS Test Series 2026 | Mock GATE 3)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (8) out of 83 users

EXAM RESPONSE EXAM STATS FEEDBACK

Aptitude

Q #1 Numerical Type Award: 1 Penalty: 0 Analytical Aptitude

If $0.75 \times 21 = 0.75 \times 5 + 0.75 \times 7 + 0.75 \times \alpha$, and $2.25 \div 9 + 1\frac{1}{4} - (6.5 \times 0.2) = \beta \times \frac{1}{4}$. Then the value of $\alpha - \beta$ is _____

Your Answer: | Correct Answer: 8.2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Fill in the blank with the most-appropriate choice given below

The waiter forgot to ___ the counter after the guests left.

A. wipe down
B. wipe away
C. wipe at
D. wipe for

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

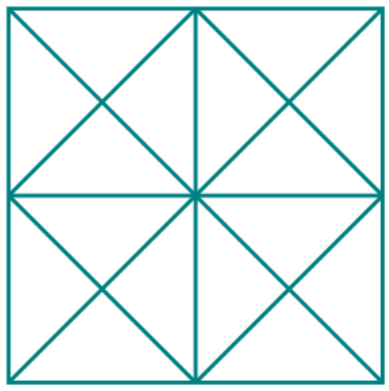
Due to Covid-19 pandemic financial crisis, Chandini had her salary reduced by 40 How much (in percentage) does she need to have her salary increased by, in order to earn as much as before?

A. 166.66%
B. 66.66%
C. 37.5%
D. 33.33%

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Spatial Aptitude

Consider the below figure. Assume that all the lines are extending on either sides of the figure. If the number of horizontal lines is α, the number of vertical lines is β, and the number of slanting lines is γ in the below figure, then the value of $3\alpha + 5\beta - 4\gamma =$ _____



Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Fill in the blank with the correct preposition.
_____ the jury's perspective, he was found guilty.

A. From
B. With
C. On
D. For

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

If $x^2 + y^2 = 72$, and $|x - y| = 6$, then the value of $(|x| + |y|)^2$ is _____

Your Answer: | Correct Answer: 108 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The sum of the algebraic series $\frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \frac{8}{1+x^8} + \frac{16}{1+x^{16}}$ is:

A. $\frac{32}{1-x^{32}}$
B. $\frac{16}{1-x^{16}}$
C. $\frac{8}{1-x^{32}}$
D. $\frac{32}{1-x^{16}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

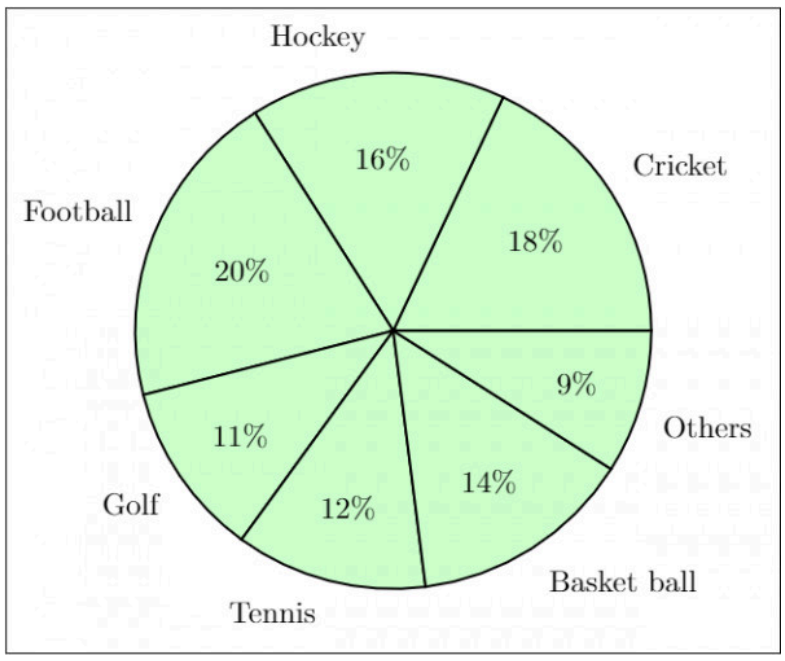
Q #8 | Numerical Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

A person is to count 12000 currency notes. Let a_n denote the number of notes he counts in the nth minute. If $a_1 = a_2 = \cdots = a_{20} = 300$ and $a_{20}, a_{21}, \ldots$ are in A.P. with common difference -4, then the minimum time taken (in minutes) by him to count all the notes is _____

Your Answer: | Correct Answer: 44 | Not Attempted | Time taken: 00min 03sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

The pie chart given below shows the spending of a country on various sports during a particular year. Study the graph carefully and answer the questions given below it.



If the total amount spent on sports during the year be Rs. $2,80,00,000$, the amount spent on Basketball exceeds that on Tennis by:

A. $Rs.\ 46,66,666$
B. $Rs.\ 5,60,000$
C. $Rs.\ 6,40,000$
D. $Rs.\ 4,44,444$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

The questions given below consists of a pair of related words followed by another pair of related words. Select the option that best expresses the relation same as the original pair.
Dearth:Scarcity::Flaw:_____

A. Replace
B. Defect
C. Blow
D. Strength

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Consider the following statements about relation R.
S_1 : The relation R on a set A is symmetric if and only if $R = R^{-1}$ where R^{-1} is the inverse relation.
S_2 : The relation R on a set A is reflexive if and only if the complementary relation $\overline{R}$ is irreflexive.
Which of the above statements is/are TRUE?

A. Only S_1
B. Only S_2
C. Both S_1 and S_2
D. Neither S_1 nor S_2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

The minimum number of states in the deterministic finite automata required to check the divisibility by 27 of a binary string $($decimal value of string being divisible by $27)$ is _____

Your Answer: | Correct Answer: 27 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

If the function $f : R \rightarrow R, f(x) = x^2$, then which of the following is correct?

A. f is surjective but not injective
B. f is both injective and surjective
C. f is injective but not surjective
D. f is neither injective nor surjective

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

If $f(w, x, y, z) = \Sigma_m(0, 2, 4, 6, 8) + \Sigma_\phi(10, 11, 12, 13, 14, 15)$, which of the followings is the minimal boolean expression for f?

A. $f(w, x, y, z) = \overline{y}$
B. $f(w, x, y, z) = \overline{z}$
C. $f(w, x, y, z) = \overline{x}$
D. $f(w, x, y, z) = \overline{w}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Numerical Type | Award: 1 | Penalty: 0 | Operating System

An operating system has 42 bits logical address. On a certain embedded device physical address space is only 16 bits. If page size is 4 KB and page table entries are of 32 bits, the size of the inverted page table (in bytes) will be _____

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Which of the following is/are the dual of the boolean expression

$f : a\,\overline{(b+\overline{c})}0 + \overline{(a+\overline{b}+c\overline{d})}1 + \overline{(\overline{a}+b+\overline{c})}0 + a\overline{b}cd + 0 + 1.$

A. $f^d(a,b,c,d) = \left(a + \overline{b \cdot \overline{c} + 1}\right) \cdot \left(\overline{a\overline{b}(c+\overline{d}) + 0}\right) \cdot \left(\overline{\overline{\overline{a}\,b\,\overline{c} + 1}}\right) \cdot (a + \overline{b} + c + d)$
B. $f^d(a,b,c,d) = 0$
C. $f^d(a,b,c,d) = \left(a + \overline{b \cdot \overline{c} + 1}\right) \cdot \left(\overline{a\overline{b}(c+\overline{d}) + 0}\right) \cdot \left(\overline{\overline{\overline{a}\,b\,\overline{c} + 1}}\right) \cdot (a + \overline{b} + c + d) \cdot (0) \cdot (1)$
D. $f^d(a,b,c,d) = \left(\overline{a\overline{b}(c+\overline{d}) + 0}\right) \cdot \left(\overline{\overline{\overline{a}\,b\,\overline{c} + 1}}\right) \cdot (a + \overline{b} + c + d) \cdot (0) \cdot (1)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

- Statement 1 : In $IPv6$ fragmentation may only be performed by routers along a packet's delivery path.
- Statement 2 : IP does not guarantee that all data will be delivered or that the data that are delivered will arrive in the proper order.

Which of the above statements is/are TRUE?

A. Both statements are True
B. Statement 1 is True but Statement 2 is False
C. Both statements are False
D. Statement 2 is True but Statement 1 is False

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following strings is a member of the language over $\{a, b\}$ defined by the regular expression $(aa + bb)^* aba (bb)^*$? (Mark all the appropriate choices)

A. $abbbba$
B. $abababb$
C. $ababb$
D. aba

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Digital Logic

Three cascaded modulus-7 counters have an overall modulus of α, and three cascaded decade counters will divide the input frequency by β. Then the value of $\alpha + \beta =$_____

Your Answer: | Correct Answer: 1343 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following commands is/are from the Hypertext Transfer Protocol (HTTP)? (Mark all the appropriate choices)

A. HEAD
B. TRACE
C. POST
D. CONNECT

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 1 | Penalty: 0 | Combinatory

How many ways are there for a horse race with three horses to finish if ties are possible? [Note: Two or three horses may tie.]

Your Answer: | Correct Answer: 13 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let P be an invertible matrix of order 3×3 and $|P| = 5$, then the value of $|\text{adj}(\text{adj}(P))|$ is _____

Your Answer: | Correct Answer: 625 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following is correct?(Mark all the appropriate choices)

A. The Domain Name System (DNS) is an application layer system that provides mapping between domain names and the associated IP addresses.
B. The DNS queries are sent as UDP packets, as a query can typically fit into a single packet.
C. For DNS system to work properly it is important that a client can have only a single server to connect to.
D. A domain name system is maintained by a distributed database system.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Let $R(A, B, C, D, E, F, G, H, I, J, K)$ be a relational schema in which the following functional dependencies are known to hold:
$\{AB \rightarrow C, BD \rightarrow EF, AD \rightarrow GH, A \rightarrow I, H \rightarrow G, I \rightarrow K, F \rightarrow H, G \rightarrow J\}$. The relational schema R is in _____

A. $1 \text{ NF and } 2 \text{ NF}$
B. $1 \text{ NF and not in } 2 \text{ NF}$
C. $3 \text{ NF and not in BCNF}$
D. $2 \text{ NF and } 3 \text{ NF}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following relational table

Student

FirstName	LastName	Age	City
Ravi	Tiwari	24	Kolkata
Anam	Raja	27	Chennai
Ranu	Mondal	38	Kolkata
Veer	Nikhil	NULL	Kolkata
Sonu	Kori	37	New Delhi
Sunil	Dipin	55	Chennai

The sum of all the values output by the following SQL query will be _____

```
select sum(age), average(age) from Student
group by City
having count(*) > 2;
```

Your Answer: | Correct Answer: 93 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

What is the minimum number of students required in a Engineering Drawing class to be sure that at least six will receive the same grade, if there are five possible grades , A, B, C, D, and F?

A. 6
B. 25
C. 26
D. 28

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

How many different one-to-one functions $f : \{0, 1, \ldots, n\} \rightarrow \{0, 1, \ldots, n+1\}$ are there?

A. ${}^{n}P_{n+2}$
B. ${}^{n+1}P_{n+2}$
C. ${}^{n+2}P_{n+1}$
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

Let $C_i (i = 0, 1, 2, \ldots n)$ be the coefficient of x^i in $(1 + x)^n$. Then $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \cdots + (-1)^n \frac{C_n}{n+2}$ is equal to

A. $\frac{1}{n+1}$
B. $\frac{1}{n+2}$
C. $\frac{1}{n(n+1)}$
D. $\frac{1}{(n+1)(n+2)}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

How many relations are there on the set $\{a, b, c, d\}$ that contain the pair (a, a)?

A. 2^{15}
B. 2^{8}
C. 2^{4}
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Which of the following is the correct first order logic representation for the sentence

$$\text{"Bill has (exactly) one sister"}$$

Here, $\text{Sisterof}(x, y) = x$ is sister of y. (Mark all the appropriate choices)

A. $\exists x(\text{Sisterof}(x, Bill) \wedge \forall y(\text{Sisterof}(y, \text{Bill}) \rightarrow (x = y)))$
B. $\forall x \forall y(\text{Sisterof}(x, \text{Bill}) \wedge \text{Sisterof}(y, \text{Bill}) \rightarrow (x = y))$
C. $\exists x(\text{Sisterof}(x, \text{Bill}) \vee \forall y(\text{Sisterof}(y, \text{Bill}) \rightarrow (x = y)))$
D. $\exists x(\text{Sisterof}(x, \text{Bill}) \wedge \neg\exists y(\text{Sisterof}(y, \text{Bill}) \wedge (x \neq y)))$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. The time complexity of the Banker's algorithm as a function of the number n of processes and m of resources is $O(n^2 \times m)$
B. Deadlock avoidance ensure that a system never enters an unsafe state
C. Banker's Algorithm is used to ensure deadlock prevention
D. Deadlock avoidance is restrictive than deadlock prevention

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let R and S be two relations on a set A. Assuming A has at least three elements, state which of the following statements is/are true .
Statement 1: If R and S are transitive then $R \cup S$ is transitive.
Statement 2 : If R and S are antisymmetric then $R \cup S$ is antisymmetric.

A. Both statements are true
B. Statement 1 is true but statement 2 is false
C. Both statements are false
D. Statement 2 is true but statement 1 is false

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

The number of two state DFAs with a designated initial state that can be constructed over the alphabet set $\Sigma = \{0, 1\}$ such that initial state is not among the set of final states is _____

Your Answer: | Correct Answer: 32 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

It is possible to apply the strength reduction transformation to realize the statement "$y = i * 21;$" where i is a loop induction variable using ____. (Mark all the appropriate choices)

A. Two adders and zero multiplier
B. One adder and zero multiplier
C. Zero adder and zero multiplier
D. Strength reduction does not apply to this statement

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

What is the output of the following C program? (Mark all the appropriate choices)

```
    #include<stdio.h>
    #include<stdlib.h>
    int main()
    {
5.      char *a="Gateoverflow"; // line 1
        char b[ ]="Gateoverflow"; // line 2
        a[3]='a'; // line 3
        b[3]='a'; // line 4
        printf("%s %s",a,b); // line 5
10. }
```

A. Gateoverflow Gateoverflow
B. Possible run time error due to line 3
C. Possible run time error due to line 4
D. Possible run time error due to line 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

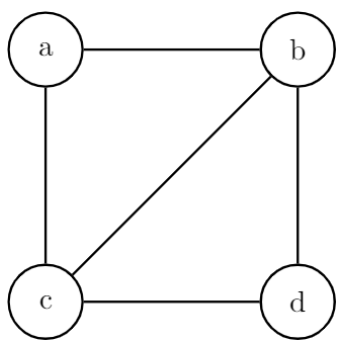
Q #26 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider a matrix Type-A such that $[A_{ij}]_{3\times 3} = X_{\text{base5}}$. The elements of matrix are also dependent on their position as, if $(i+j)\%2 = 0$, then A_{ij} is even and else A_{ij} is odd. Calculate the number of non-symmetric matrices of Type-A _____

Your Answer: | Correct Answer: 3564 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

How many walks of length four are there from a to c in the simple graph G?



Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What is the output of the following program?
(Assume that the value returned by the malloc call is 1001.)

```c
#include<stdio.h>
#include<stdlib.h>
void fun(int **y);
int main()
{
   int *x = (int*)malloc(sizeof(int));
   *x = 132;
   fun(&x);
   printf("%d %u\n",*x,x);
}
void fun(int **y)
{
   int z=21;
   **y = z;
   printf("%u ",*y);
}
```

A. $21\ 132\ 1001$
B. $1001\ 21\ 1001$
C. 1001 garbage 1001
D. $21\ 21\ 132$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the grammar

- $S \rightarrow AaAb \mid BbBa$
- $A \rightarrow \varepsilon$
- $B \rightarrow \varepsilon$

If the number of shift-reduce conflicts for $LR(0)$ parsing of the above grammar is α, and the number of reduce-reduce conflicts is β, then the value of $\alpha - \beta =$_____

Your Answer: | Correct Answer: -1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following pseudocode:

```
struct node {
        int data;
        struct node* left;
        struct node* right;
};
struct node* Dosomething(struct node* root, int key)
{
        struct node* temp;
        if(root == NULL)
            return root;
        if(root -> data == key)
            return root;
    if( key < root -> data)
    {
        if(root -> left == NULL)
            return root;
        temp = Dosomething(root -> left,key);
        if( abs(temp -> data - key) > abs(root -> data - key))
            return root;
        else if( abs(temp -> data - key) < abs(root -> data - key))
            return temp;
     }
   else
   {
        if(root -> right==NULL)
             return root;
        temp = Dosomething(root -> right,key);
        if( abs(temp -> data - key) > abs(root -> data - key))
             return root;
        else if( abs(temp data - key) < abs(root -> data - key))
             return temp;
   }
}
```

A. The element which is closest to the given key
B. The inorder successor of the key
C. The inorder predecessor of the key
D. May sometimes give garbage value

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 Numerical Type | Award: 2 | Penalty: 0 | Data Structures

AVL tree is a self-balancing Binary Search Tree (BST) where the difference between heights of left and right subtrees cannot be more than one for all nodes. What is the different shapes possible for an AVL tree with minimum number of nodes and height 4? (Assume height of root as 0).

Your Answer: | Correct Answer: 128 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 Numerical Type | Award: 2 | Penalty: 0 | Data Structures

The preorder traversal sequence of a binary search tree is $12, 4, 2, 1, 3, 8, 6, 10, 17, 15, 14, 16, 19, 18, 20$. If the sum of the value of nodes whose index values (position from beginning) of preorder and postorder traversals are same is k, then the value of k is _____

Your Answer: | Correct Answer: 44 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The number of canonical covers of the set of functional dependencies : $\{A \rightarrow BC, B \rightarrow CA, C \rightarrow AB\}$ is _____

A. 1
B. 2
C. 3
D. > 3

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a main memory with four-page frames and the following sequence of page references:

$$1, 2, 3, 4, 1, 2, 5, 4, 3, 2, 1, 4, 5$$

Which one of the following is true with respect to page replacement policies First In First Out (FIFO) and Least Recently Used (LRU)?

A. Both incur the same number of page faults
B. FIFO incurs 2 more page faults than LRU
C. LRU incurs 2 more page faults than FIFO
D. FIFO incurs 1 more page faults than LRU

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Numerical Type | Award: 2 | Penalty: 0 | Probability

A delivery company divides their packages into weight classes. Suppose packages in the 13 to 21 kilogram class are uniformly distributed, meaning that all weights within that class are equally likely to occur. If the probability that a package weighs between 14 and 16.5 kilograms is α, the probability that a package weighs less than 17 kilograms is β, the probability that a package weighs at least 16 kilograms is γ, and the probability that a package weighs at most 19 kilograms is ρ. Then the value of $\alpha - \beta + \gamma - \rho$ is _____ (upto two decimal places)

Your Answer: | Correct Answer: -0.32:-0.31 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

The number of all the positive integers x less than 29, such that $x^{86} \equiv 6 \mod 29$, is _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Free disk space can be kept track of using a free list or a bitmap. Disk addresses require D bits. For a disk with B blocks and D having the value 32 bits, the percentage of the disk space that must be free for free list to use same memory as bitmap will be _____ (up to three decimal place).

Your Answer: | Correct Answer: 3.125 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

The following elements $\{0, 64, 25, 49, 4, 19, 42, 5, 40, 20\}$ are inserted in an empty hash table using the hash function $\mod 10$ and separate chaining technique for collision avoidance. Let n be the number of empty slots and m be the number of non empty slots after the insertions. Three more keys are inserted in alternate even odd pattern independently starting with an even key. What is the probability that after these 3 insertions the ratio of n and m will remain the same? Give answer up to 3 decimal places.

Your Answer: | Correct Answer: 0.144:0.145 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 Numerical Type | Award: 2 | Penalty: 0 | Programming in C

What will be the output when the following C program is executed?

```c
#include<stdio.h>
#define N 100000
int foo(int a[], int *b)
{
    a[0] = 1;
    a[1] = 2;
    b++;
    b[0] = 3;
    b[1] = 4;
    return a[0];
}
int main()
{
    int a[N];
    for(int i = 0; i < N; i++)
    {
    a[i] = i;
    }
    a[0] = foo(a, a);
    printf("%d", a[0]+a[1]+a[2]+a[3]);
    return 0;
}
```

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A device with transfer rate of 25 KBps is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 12 microseconds. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating device under interrupt driven mode?

Your Answer: | Correct Answer: 3.3:3.4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output when the following C program is executed?

```c
#include<stdio.h>
#include<string.h>
int main()
{
    char s[] = "sayonara";
    for(int i = 0; i < strlen(s); i++)
    {
        s[i] <= 'm'? s[i++]++ : s[i]+2;
    }
    printf("%s",s);
    return 0;
}
```

A. ubyqpbrb
B. The code can produce a runtime error
C. Compilation error
D. sbyonbrb

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the language L over $\Sigma = \{0\}$ consisting of strings of length a power of 2. You are using the pumping lemma to show that L is not regular. You begin by supposing L were accepted by a DFA with k states. What would be a good choice of a string xyz that would allow you to obtain a contradiction?

A. 0^{2k}
B. $0^{(2^k)}0^k$
C. 0^{2^k}
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a secondary storage system of size $8\ \text{TB}$, with 2048-byte sized blocks. Assume that the file system uses a multilevel inode data structure to track data blocks of a file. The inode has $128\ \text{bytes}$ of space available to store pointers to data blocks, including two indirect blocks, a single double indirect block, and several direct blocks. What is the maximum file size that can be stored in such a file system in kilobytes?

A. 327680
B. 16526
C. 526394
D. 65536

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a five-stage pipeline with ideal CPI. Assume that 25% of the instructions executed are branch instructions out of which 70% are taken branches. The pipeline architecture is such that there is no penalty for a non-taken branch whereas every taken branch incurs on average a penalty of 1 cycle. What will be the effective pipeline speedup?

A. 4.44

B. 4.59
C. 4.25
D. 4.66

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a DMA controller transferring 32 bit words to memory using cycle stealing. The words are assembled from a device that transmits bytes at a rate of 24 Mbytes per second. The CPU fetches and executes instructions at an average rate of 2 billion instructions per second. By how much time will the CPU be slowed down because of the DMA transfer?

A. 0.6%
B. 0.12%
C. 0.3%
D. 0.10%

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Given $f(a,b,c,d) = ab\overline{c} + \overline{c}d + \overline{a}c\overline{d} + \overline{b}c\overline{d}$, and $g(a,b,c,d) = (a+b+\overline{c}+\overline{d})(\overline{b}+\overline{c}+d)(\overline{a}+b+\overline{d})$.

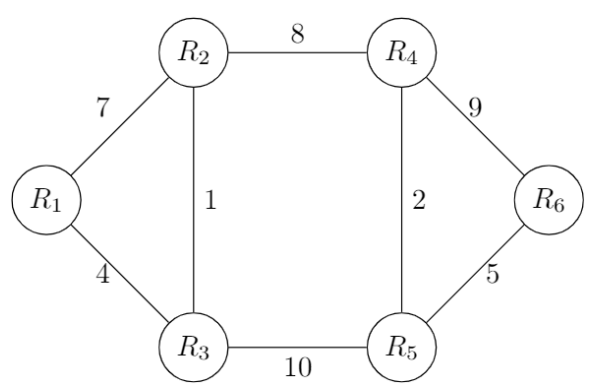
Which of the following is/are correct? (Mark all the appropriate choices)

A. $(f \cdot g)(a,b,c,d) = \Sigma_m(1,2,5,10,12,13)$
B. $(f + g)(a,b,c,d) = \Sigma_m(0,1,2,4,5,6,7,8,9,10,12,13,15)$
C. $(f \oplus g)(a,b,c,d) = \Sigma_m(0,4,6,7,8,9,15)$
D. $(f \odot g)(a,b,c,d) = \Sigma_m(1,2,3,5,10,11,12,13,14)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider a network with 6 routers R_1 to R_6 connected with links having weights as shown in the following diagram.

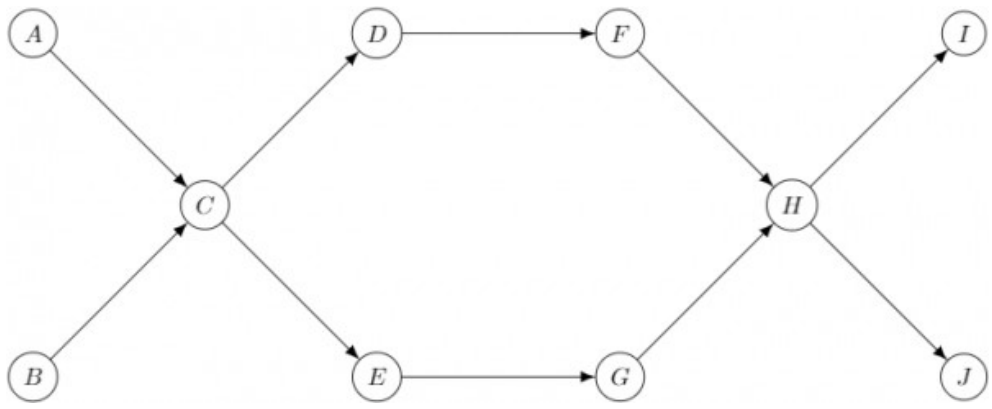


All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

The number of topological-orderings (sortings) of the graph given is _____



Your Answer: | Correct Answer: 24 | Not Attempted | Time taken: 00min 00sec | Discuss

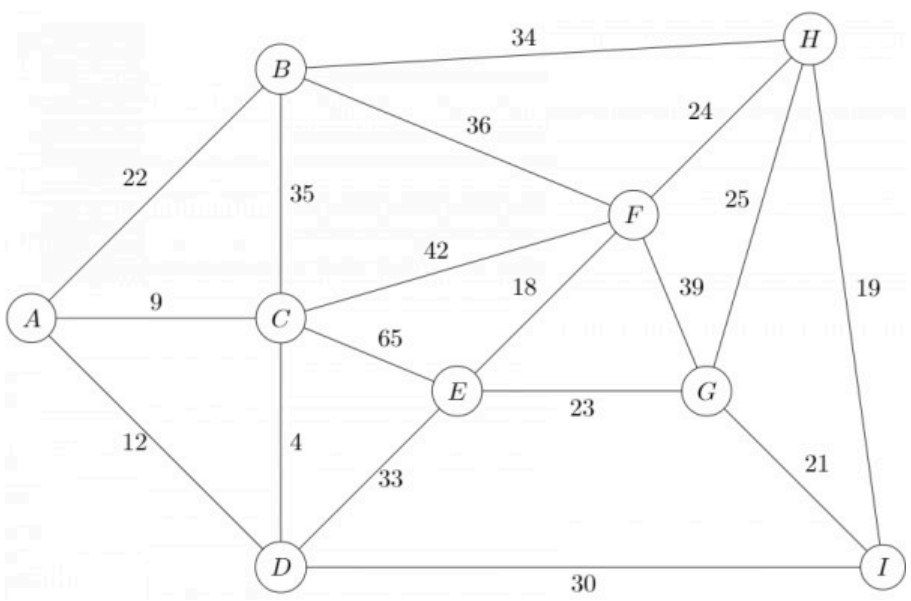
Q #49 Numerical Type | Award: 2 | Penalty: 0 | Calculus

If the function $f(x) = \begin{cases} \alpha\sqrt{x+1} & ; 0 \le x \le 3 \\ \beta x + 2 & ; 3 < x \le 5 \end{cases}$ is differentiable, then the value of $\alpha - \beta$ is _____

Your Answer: | Correct Answer: 1.2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

If the minimum spanning tree for the graph below has total weight α, and the difference between the maximum and minimum weights in it is β, then the value of $\alpha + \beta$ is _____



Your Answer: | Correct Answer: 172 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Two coins are in a box. The coins look alike, but one coin is fair (with equal probabilities for Head and Tail), while the other coin is biased, with $2/3$ probability of Heads. One of the coins is randomly chosen from the hat, without knowing which of the two it is. Now this chosen coin is tossed twice, showing Heads both times. Given this information, what is the probability that the chosen coin is the fair coin?

A. $\frac{4}{5}$
B. 0.692
C. $\frac{9}{25}$
D. $\frac{1}{2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are NOT regular? (Mark all the appropriate choices)

A. $L = \{xww_R \mid w, x \in (a+b)^+\}$
B. $L = \{xww \mid w, x \in (a+b)^*\}$
C. $L = \{w \mid w \in a+b)^*, w \text{ contains an equal number of occurrences of the substrings } 01 \text{ and } 10\}$
D. $L = \{wxw_R \mid w, x \in (a+b)^+\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume a main memory access time of $112\ ns$ and a memory system capable of a sustained transfer rate of 8 GBps. If the block size is 64 bytes, what is the maximum number of outstanding misses we need to support assuming that we can maintain the peak bandwidth given the request stream and that accesses never conflict. For simplicity, ignore the time between misses.

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following Syntax Directed Translation Scheme, with non-terminals $\{S, T, F\}$ and terminals $\{\text{num}\}$. $S \rightarrow T\#S \quad \{S.val = T.val - S.val;\}$
$\quad |T \quad \{S.val = T.val;\}$

$T \rightarrow F * T \quad \{T.val = F.val * T.val;\}$
$\quad |F \quad \{T.val = F.val;\}$ $\qquad F \rightarrow \text{num} \quad \{F.val = num.lval;\}$

The value of $5\#3 * 11 * 7\#3\#2 * 4\#2$ using the syntax directed translation scheme described by the above rules is _____

Your Answer: | Correct Answer: -229 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A disk has 4 equidistant tracks. The diameters of the innermost and outermost tracks are $1\ \text{cm}$ and $6\ \text{cm}$ respectively. The innermost track has a storage capacity of $16\ \text{MB}$. If the disk has 24 sectors per track and is currently at the end of the 3rd sector of the inner-most track and the head can move at a speed of

12 meters/sec and it is rotating at constant angular velocity of 7200 RPM, the time (in milliseconds) it will take to read 10 MB contiguous data starting from the sector 4 of the outer-most track will be _____

Your Answer: | **Correct Answer: 13.4:13.8** | **Not Attempted** | **Time taken: 00min 00sec** | **Discuss**

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